INCH-POUND
MIL-DTL-24707/1A
5 December 2017
SUPERSEDING
MIL-C-24707/1
27 January 1989

DETAIL SPECIFICATION SHEET

CASTINGS, FERROUS, FOR MACHINERY AND STRUCTURAL APPLICATIONS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-C-24707.

1. SCOPE

1.1 Scope. This specification covers steel castings for machinery and structural applications below 775 degrees Fahrenheit (°F) where impact strength may be a consideration.

1.2 Classification. Castings are of the following grades, as specified (see 6.2).

Grade	A1Q of ASTM A757/A757M	Grade	E2Q of ASTM A757/A757M
	A2Q of ASTM A757/A757M		WCA of ASTM A216/A216M
	C1Q of ASTM A757/A757M		WCB of ASTM A216/A216M
	E1Q of ASTM A757/A757M		WCC of ASTM A216/A216M
	E2N of ASTM A757/A757M		4B of ASTM A487/A487M
			14A of ASTM A487/A487M

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of this document are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-C-24707 - Castings, Ferrous, General Specification for

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A216/A216M - Standard Specification for Steel Castings, Carbon, Suitable for Fusion Welding, for High-Temperature Service

ASTM A487/A487M - Standard Specification for Steel Castings Suitable for Pressure Service

ASTM A757/A757M - Standard Specification for Steel Castings, Ferritic and Martensitic, for Pressure-Containing and Other Applications, for Low-Temperature Service

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with MIL-C-24707.

3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 General. The requirements for castings covered by this specification shall be in accordance with MIL-C-24707, ASTM A216/A216M, ASTM A487/A487M, or ASTM A757/A757M, and as specified herein.

3.4 Supplementary requirements. When specified (see 6.2), additional supplementary requirements shall be met.

3.5 Chemical analysis. Chemical analysis shall meet the requirements in the specified commercial specification. In addition, for carbon steels, the specified residual elements shall be determined.

3.6 Impact requirements. Unless otherwise specified (see 6.2), the Charpy V-notch impact requirements shall be those listed in the specified commercial specification. When no requirements are listed in the commercial specifications, they shall be 20 foot-pounds at 10 °F (-12 degrees Celsius [°C]) except when ordered for deck applications (see 6.2). Deck application requirements shall be 20 foot-pounds at -20 °F (-29 °C).

3.7 Stress relief heat treatment. When specified (see 6.2), the stress relieving temperature shall be at least 50 °F (30 °C) but not more than 100 °F (60 °C) below the tempering temperature. The mechanical properties shall be determined after the stress relief heat treatment.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

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4.2 First article inspection. First article inspection shall be performed when a first article sample is required (see 3.1). This inspection shall be in accordance with MIL-C-24707.

4.3 Conformance inspection. Conformance inspection shall be in accordance with MIL-C-24707.

4.4 Sampling. Unless otherwise specified (see 6.2), the method of sampling shall be in accordance with MIL-C-24707.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. These castings are intended for structural and machinery applications below 775 °F where impact strength may be a consideration. Castings procured to this specification are form, fit, and function replacements for prior revisions.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Grade of steel required (see 1.2).
c. When first article is required (see 3.1).
d. If any additional supplementary requirements of ASTM A216/A216M, ASTM A487/A487M, ASTM A757/A757M, or other requirements are required (see3.4).
e. If castings are for deck applications (see3.6).
f. If other impact requirements are to be specified (see3.6).
g. If a stress relief heat treatment is required (see3.7).
h. If sampling is other than specified (see4.4).
i. Packaging requirements (5.1).

6.3 Subject term (key word) listing.

Impact strength

Steel

Stress relief heat treatment

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army – MR
Navy – SH
Air Force – 20

Review activity:
Army – CR4

Preparing activity:
Navy – SH
(Project MECA-2017-002)

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